Regression Problems (Correlation, Independence, Rank-Based Regression)

1 The Independence Problem: Correlation and Regression

1.1 Introduction

One of the most fundamental questions in statistics is:

Are two variables related, or are they independent?

This question appears throughout applications:

- Does higher blood pressure relate to higher risk of stroke?
- Does income level relate to educational attainment?
- Does tumor size relate to survival time?

To study such relationships, we use two closely connected tools:

- **Correlation:** measures the strength of association between two variables.
- **Regression:** models how one variable changes as another variable changes.

Key distinction:

- Correlation is symmetric: it treats X and Y equally.
- Regression is directional: it models a response Y given predictors X.

Regression relationships can take many forms:

- Simple linear regression ($Y = \beta_0 + \beta_1 X + \varepsilon$),
- Multiple regression ($Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$),
- Nonlinear regression (polynomial, logistic, generalized models).

The goal of this lecture is to understand how correlation and regression are mathematically connected, especially through Pearson correlation.

1.2 Association: discrete vs continuous

Consider a random pair (X, Y) and an i.i.d. sample

$$(X_1, Y_1), \ldots, (X_n, Y_n).$$

The appropriate measure of association depends on whether the variables are **discrete** or **continuous**.

Discrete case

If X and Y are discrete (categorical), dependence is often studied using:

- contingency tables,
- odds ratios,
- chi-square (χ^2) tests.

Independence means:

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y) \quad \text{for all possible levels of } x, y.$$

Thus, knowing X gives no information about Y.

Continuous case

If X and Y are continuous, dependence is often measured using:

- Pearson correlation (linear association),
- Spearman correlation (rank-based monotone association),
- Kendall correlation (sign-based concordance association).

These methods do not rely on contingency tables and are better suited for quantitative variables.

2 Pearson's Correlation Coefficient

2.1 Definition

Let X and Y be random variables with finite means μ_X, μ_Y and finite standard deviations σ_X, σ_Y.

Pearson's population correlation coefficient is defined as

$$\rho = \frac{\mathbb{E}\{(X - \mu_X)(Y - \mu_Y)\}}{\sigma_X \sigma_Y}.$$

The numerator measures how X and Y vary together, while the denominator rescales the covariance so that the coefficient is unit-free and lies in $[-1, 1]$.

Equivalently,

$$\rho = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)}{\sigma_X \sigma_Y}.$$

Thus, correlation is simply the covariance standardized by the marginal standard deviations.

Range and Interpretation

It always holds that

$$-1 \le \rho \le 1.$$

- $\rho = 1$: perfect increasing linear relationship (all points lie on a line with positive slope),
- $\rho = -1$: perfect decreasing linear relationship,
- $\rho = 0$: no linear association.

Correlation and Independence

If X and Y are independent, then

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y),$$

Why does independence imply zero covariance?
If X and Y are independent, then their joint density factorizes:

$$f_{X,Y}(x,y) = f_X(x)\, f_Y(y).$$

Hence,

$$\mathbb{E}(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f_{X,Y}(x,y)\,dx\,dy = \int\int xy\, f_X(x)\, f_Y(y)\,dx\,dy.$$

Since the integrand separates into a function of x times a function of y, we may write

$$\mathbb{E}(XY) = \left(\int x f_X(x)\,dx\right)\left(\int y f_Y(y)\,dy\right).$$

Therefore,

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y).$$

Consequently,

$$\mathrm{Cov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = 0.$$

which implies $\rho = 0$.

Important: The converse is not generally true.

$$\rho = 0 \not\Rightarrow X \perp Y.$$

For example, if $X \sim N(0,1)$ and $Y = X^2$, then X and Y are dependent but $\mathrm{Cov}(X,Y) = 0$. However, under **bivariate normality**,

$$\rho = 0 \quad \Longleftrightarrow \quad X \perp Y.$$

Proof. $(\Rightarrow)$ Assume $\rho = 0$. Consider the centered variables

$$X_c = X - \mu_X, \qquad Y_c = Y - \mu_Y.$$

Because (X,Y) is bivariate normal, (X_c, Y_c) is also bivariate normal with mean 0 and covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \mathrm{Cov}(X,Y) \\ \mathrm{Cov}(X,Y) & \sigma_Y^2 \end{pmatrix} = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}.$$

When $\rho = 0$, this becomes diagonal:

$$\Sigma = \begin{pmatrix} \sigma_X^2 & 0 \\ 0 & \sigma_Y^2 \end{pmatrix}.$$

Let (X, Y) be bivariate normal with means μ_X, μ_Y and standard deviations $\sigma_X, \sigma_Y > 0$. The joint density is

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{ -\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho \left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 \right] \right\}.$$

Assume $\rho = 0$. Then $\sqrt{1-\rho^2} = 1$ and the cross-term vanishes, yielding

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y} \exp\left\{ -\frac{1}{2} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 \right] \right\}.$$

Now split the exponential into a product:

$$f_{X,Y}(x, y) = \left[\frac{1}{\sqrt{2\pi}\sigma_X} \exp\left\{ -\frac{1}{2} \left(\frac{x-\mu_X}{\sigma_X}\right)^2 \right\} \right] \left[\frac{1}{\sqrt{2\pi}\sigma_Y} \exp\left\{ -\frac{1}{2} \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 \right\} \right].$$

But the bracketed terms are exactly the marginal normal densities:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} \exp\left\{ -\frac{1}{2} \left(\frac{x-\mu_X}{\sigma_X}\right)^2 \right\}, \qquad f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma_Y} \exp\left\{ -\frac{1}{2} \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 \right\}.$$

Hence, when $\rho = 0$,

$$f_{X,Y}(x, y) = f_X(x)\, f_Y(y) \quad \text{for all } (x, y),$$

which is the defining condition for independence. Therefore $X \perp Y$.

This proves $\rho = 0 \Rightarrow X \perp Y$ for bivariate normal. The converse ($X \perp Y \Rightarrow \rho = 0$) is immediate, completing the proof.

<u>(⇐)</u> If $X \perp Y$, then

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y),$$

so $\mathrm{Cov}(X, Y) = 0$, hence $\rho = \mathrm{Cov}(X, Y)/(\sigma_X\sigma_Y) = 0$.

Combining the two directions proves the equivalence. □

2.2 Sample Pearson Correlation

Given observed data (X_i, Y_i), $i = 1, \ldots, n$, the sample correlation is

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}\,\sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}.$$

Equivalently,

$$r = \frac{\widehat{\mathrm{Cov}}(X, Y)}{\hat{\sigma}_X\hat{\sigma}_Y}.$$

It is unit-free and satisfies $-1 \le r \le 1$.

2.3 Relation to Simple Linear Regression

Consider the least squares regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

The least squares slope estimator satisfies

$$\hat{\beta}_1 = \frac{\widehat{\mathrm{Cov}}(X, Y)}{\widehat{\mathrm{Var}}(X)} = r \left(\frac{\hat{\sigma}_Y}{\hat{\sigma}_X} \right).$$

Thus:

- The sign of $\hat{\beta}_1$ is determined by r,
- The magnitude of r measures strength of linear association,
- Regression rescales correlation into the units of Y.

2.4 Testing and Inference

- Under bivariate normality, testing $H_0 : \rho = 0$ uses

$$t = r\sqrt{\frac{n-2}{1-r^2}}, \qquad t \sim t_{n-2}.$$

- A permutation test is distribution-free: shuffle Y relative to X and recompute r.
- A bootstrap confidence interval can be obtained by resampling pairs

$$(X_i, Y_i) \quad \text{with replacement.}$$

- Pearson correlation is sensitive to outliers and requires finite variance. For example, the Cauchy distribution does not have a finite standard deviation, so ρ is not defined.

The connection between correlation and regression coefficient does not hold in the case of multiple linear regression.

2.5 Partial Correlation in Multiple Regression

In multiple regression, we often wish to measure the linear association between Y and one predictor X_1, after adjusting for other predictors $X_2, \ldots, X_p$. This leads to the notion of **partial correlation**.

Consider the multiple regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_p X_{pi} + \varepsilon_i.$$

The coefficient β_1 measures the effect of X_1 holding all other predictors fixed.

Definition (Residual Interpretation)

Let:

- $\{e_{Y,i}\}_{i=1}^{n}$ be the residuals from regressing Y on $X_2, \ldots, X_p$,
- $\{e_{X_1,i}\}_{i=1}^{n}$ be the residuals from regressing X_1 on $X_2, \ldots, X_p$.

The **sample partial correlation** between Y and X_1 given $X_2, \ldots, X_p$ is

$$r_{Y,X_1|X_2,\ldots,X_p} = \mathrm{Cor}(e_Y, e_{X_1}).$$

Thus, partial correlation is simply the ordinary correlation between the parts of Y and X_1 that remain after removing their linear dependence on the other predictors.

Connection to Multiple Regression Coefficient

In the multiple regression model, the slope $\hat{\beta}_1$ satisfies

$$\hat{\beta}_1 = r_{Y,X_1|X_2,\ldots,X_p} \, \frac{\hat{\sigma}_{Y|X_{2:p}}}{\hat{\sigma}_{X_1|X_{2:p}}},$$

where $\hat{\sigma}_{Y|X_{2:p}}$ and $\hat{\sigma}_{X_1|X_{2:p}}$ are residual standard deviations.
Thus:

- The sign of $\hat{\beta}_1$ is determined by the partial correlation.
- The magnitude of $\hat{\beta}_1$ rescales the partial correlation.
- Partial correlation measures the unique linear contribution of X_1 to explaining Y.

Testing via Partial Correlation

Testing

$$H_0: \ \beta_1 = 0$$

in multiple regression is equivalent to testing

$$H_0: \ r_{Y,X_1|X_2,\ldots,X_p} = 0.$$

The corresponding t-statistic is

$$t = r_{Y,X_1|X_2,\ldots,X_p} \sqrt{\frac{n-p-1}{1 - r^2_{Y,X_1|X_2,\ldots,X_p}}}, \qquad t \sim t_{n-p-1}.$$

This mirrors the simple regression result, but with degrees of freedom reduced by the number of predictors.

Matrix Form (Precision Matrix Interpretation)

Let $\boldsymbol{\Sigma}$ be the population covariance matrix of $(Y, X_1, \ldots, X_p)$, and let $\boldsymbol{\Omega} = \boldsymbol{\Sigma}^{-1}$ denote the precision matrix.

Then the population partial correlation between Y and X_1 given the other predictors is

$$\rho_{Y,X_1|X_2,\ldots,X_p} = -\frac{\Omega_{Y,X_1}}{\sqrt{\Omega_{Y,Y}\,\Omega_{X_1,X_1}}}.$$

Thus:

- Zero partial correlation corresponds to $\Omega_{Y,X_1} = 0$,
- Under multivariate normality, zero partial correlation implies conditional independence.

Interpretation

- Ordinary correlation measures marginal association.
- Partial correlation measures association after adjustment.
- In regression, slopes correspond to scaled partial correlations.

3 Independence Tests Based on Signs: Kendall's τ

3.1 Independence Hypothesis

Given i.i.d. observations (X_i, Y_i) from $F_{X,Y}$, we test

$$H_0: \; F_{X,Y}(x, y) = F_X(x)F_Y(y) \quad \text{(independence)}$$

versus H_1 : dependence.

Kendall's τ detects <u>monotone association</u> rather than purely linear association, but also not any association.

3.2 Population Kendall Correlation

Let (X_1, Y_1) and (X_2, Y_2) be two independent observations from the joint distribution of (X, Y).

Kendall's population coefficient is defined as

$$\tau = 2\mathbb{P}\{(Y_2 - Y_1)(X_2 - X_1) > 0\} - 1.$$

What does this probability mean?

For a randomly chosen pair of observations:

- The pair is called **concordant** if the ordering of X and the ordering of Y agree:

$$(X_2 - X_1)(Y_2 - Y_1) > 0.$$

- The pair is called **discordant** if the orderings disagree:

$$(X_2 - X_1)(Y_2 - Y_1) < 0.$$

Let, $P(\text{concordant})$ is the probability that, for two randomly chosen observation pairs (X_i, Y_i) and (X_j, Y_j), the one with the larger X value also has the larger Y value. And $P(\text{discordant})$ is the probability that, for two randomly selected observations, the one with the larger X value has the smaller Y value.

(If ties occur, the product equals 0; in continuous distributions, ties occur with probability 0.)

Thus,

$$\tau = P(\text{concordant}) - P(\text{discordant}).$$

Since

$$P(\text{concordant}) + P(\text{discordant}) = 1 \quad \text{(in the continuous case)},$$

we may also write

$$\tau = 2P(\text{concordant}) - 1.$$

Interpretation

- $\tau = 1$: all pairs are concordant (perfect increasing monotone relationship),
- $\tau = -1$: all pairs are discordant (perfect decreasing monotone relationship),
- $\tau = 0$: concordant and discordant pairs occur equally often.

Therefore, Kendall's τ measures the tendency of the relative ordering of X to agree with the relative ordering of Y. It captures monotone association rather than strictly linear association.

3.3 Concordant and Discordant Pairs

For each pair (i, j) with $i < j$, define

$$Q\big((X_i, Y_i), (X_j, Y_j)\big) = \begin{cases} 1, & \text{if } (X_j - X_i)(Y_j - Y_i) > 0, \\ -1, & \text{if } (X_j - X_i)(Y_j - Y_i) < 0, \\ 0, & \text{if } (X_j - X_i)(Y_j - Y_i) = 0 \text{ (tie)}. \end{cases}$$

The Kendall sign statistic is

$$K = \sum_{1 \le i < j \le n} Q\big((X_i, Y_i), (X_j, Y_j)\big).$$

Since there are

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

distinct pairs, the normalized estimator of Kendall's population coefficient is

$$\hat{\tau} = \frac{K}{\binom{n}{2}} = \frac{2K}{n(n-1)}.$$

Thus $\hat{\tau}$ is simply the difference between the proportion of concordant pairs and the proportion of discordant pairs.

3.4 Large-Sample Approximation

Under the null hypothesis of independence,

$$H_0: \ X \perp Y,$$

concordant and discordant pairs occur with equal probability. Hence,

$$\mathbb{E}(K) = 0.$$

It can be shown (for continuous distributions, i.e. no ties) that

$$\mathrm{Var}(K) = \frac{n(n-1)(2n+5)}{18}.$$

(Proof for variance is not trivial. One can either use results from U-statistic theory or enumerate the full combinatorial space. The somewhat complicated variance formula arises from dependence between overlapping index pairs.)

Therefore, the standardized statistic

$$K^* = \frac{K}{\sqrt{n(n-1)(2n+5)/18}}$$

has approximately a standard normal distribution for large n:

$$K^* \ \xrightarrow{d} \ N(0,1).$$

Interpretation:

- If H_0 is true, K^* fluctuates around 0.
- Large positive values indicate positive monotone association.
- Large negative values indicate a negative monotone association.

Thus Kendall's test is asymptotically normal and may be used to construct large-sample hypothesis tests and confidence intervals.

3.5 Kendall's Statistic as a U-Statistic

The Kendall sign statistic

$$K = \sum_{1 \le i < j \le n} h(Z_i, Z_j), \qquad Z_i = (X_i, Y_i),$$

can be written using the symmetric kernel

$$h(Z_i, Z_j) = \mathrm{sign}\big((X_j - X_i)(Y_j - Y_i)\big).$$

Thus the normalized estimator

$$\hat{\tau} = \frac{2}{n(n-1)} K = \frac{1}{\binom{n}{2}} \sum_{i<j} h(Z_i, Z_j)$$

is a **U-statistic of order 2**.

General U-statistic theory:

For a symmetric kernel $h(Z_1, Z_2)$ with finite variance,

$$U_n = \frac{1}{\binom{n}{2}} \sum_{i<j} h(Z_i, Z_j).$$

4 Independence Tests Based on Ranks: Spearman

4.1 Motivation

Pearson correlation measures linear association. However, in many practical situations:

- The relationship may be monotone but nonlinear (e.g. exponential growth),
- The data may contain outliers that distort covariance,
- The variables may have heavy-tailed distributions,
- Variances may be infinite (e.g. Cauchy-type behavior).

Since Pearson correlation depends directly on means and variances, it can be unstable under these conditions.

Rank-based measures replace raw values by their orderings. Because order is invariant under monotone transformations, rank-based correlations detect general monotone relationships and are less sensitive to extreme observations.

4.2 Definition of Spearman's Rank Correlation

Spearman's population coefficient ρ_s measures monotone association between two variables.

Given data (X_i, Y_i):

- Replace X_i by its rank R_i among $X_1, \ldots, X_n$,
- Replace Y_i by its rank S_i,
- Compute the ordinary Pearson correlation of (R_i, S_i).

Thus the sample Spearman coefficient is

$$r_s = \mathrm{Cor}(R, S).$$

Equivalently, it is the Pearson correlation applied to the empirical distribution functions of X and Y.

Because ranks are invariant under strictly increasing transformations, r_s measures the association between the transformed variables $F_X(X)$ and $F_Y(Y)$ (the marginal cumulative distribution functions (CDFs) of X and Y, respectively).

4.3 Interpretation

- $r_s \approx 1$: strong increasing monotone relationship,
- $r_s \approx -1$: strong decreasing monotone relationship,
- $r_s \approx 0$: no monotone association.

Unlike Pearson correlation, Spearman does not require linearity. It captures any strictly monotone relationship.

Special Case: No Ties

If there are no ties, Spearman's coefficient simplifies to

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}, \qquad d_i = R_i - S_i.$$

This formula shows that Spearman correlation is essentially a normalized measure of the squared distance between the two rankings.

If the rankings are identical ($d_i = 0$ for all i), then $r_s = 1$. If one ranking is exactly reversed, then $r_s = -1$.

4.4 Comparison: Spearman vs Kendall

Both Spearman's ρ_s and Kendall's τ measure monotone association, but they differ conceptually and mathematically.

1. Definition

- Spearman:

$$\rho_s = \mathrm{Cor}(\text{ranks of } X,\ \text{ranks of } Y).$$

- Kendall:

$$\tau = P(\text{concordant}) - P(\text{discordant}).$$

Thus:

- Spearman measures rank covariance,
- Kendall measures pairwise ordering agreement.

2. Interpretation

- τ has a direct probabilistic meaning: it equals the difference between the probability of concordance and discordance.
- ρ_s measures how close two entire rankings are in a global squared-distance sense.

3. Robustness

Both are robust compared to Pearson.

- Kendall's τ is slightly more robust because it depends only on pairwise signs.
- Spearman's ρ_s uses rank magnitudes and can be slightly more sensitive to extreme rank discrepancies.

4. Asymptotic Efficiency

Under bivariate normality:

- Kendall's τ has asymptotic relative efficiency (ARE) approximately 0.91 relative to Pearson.
- Spearman's ρ_s has ARE approximately 0.95 relative to Pearson.

Thus both lose little efficiency when the true relationship is linear normal, but gain robustness under deviations from normality.

5. Mathematical Structure

- Kendall's statistic is a U-statistic of order 2.
- Spearman's statistic can be expressed as a U-statistic of order 3 (or as a correlation of empirical distribution transforms).

This explains why both are asymptotically normal.

4.5 Summary

Measure	Detects	Interpretation	Robustness
Pearson	Linear	Covariance-based	Low
Spearman	Monotone	Rank correlation	Moderate
Kendall	Monotone	Probability of concordance	High

In practice:

- Use Pearson when linearity and normality are plausible.
- Use Spearman for general monotone association.
- Use Kendall when a probabilistic interpretation and stronger robustness are desired.

4.6 Independence Interpretation

Under independence, the ranks of X and Y behave like independent random permutations.

Hence,

$$H_0 : \; X \perp Y \quad \Longrightarrow \quad \rho_s = 0.$$

Spearman is effective for detecting monotone dependence.

Efficiency remark:

- If the true relationship is linear and normal, Pearson is most efficient.
- If the relationship is monotone but nonlinear, Spearman and Kendall often perform better.
- Rank-based methods are more robust to outliers.

5 A Distribution-Free Test for Independence Against Broad Alternatives (Hoeffding, 1948)

5.1 Motivation

Many classical independence tests (e.g. Pearson correlation) are designed to detect linear association.

Rank-based measures such as Kendall's τ and Spearman's ρ_s detect monotone dependence.

However, dependence can be:

- nonlinear and non-monotone,
- symmetric but curved,
- oscillatory,
- or otherwise structured.

Hoeffding (1948) proposed a distribution-free test that detects any departure from independence, not just monotone relationships.

5.2 Population Quantity

Let (X, Y) have joint distribution function $F(x, y)$ and marginal distributions $F_X(x)$ and $F_Y(y)$.

Under independence,

$$F(x, y) = F_X(x)F_Y(y).$$

Hoeffding defined a measure of dependence based on the squared distance between these two quantities:

$$D = \int [F(x, y) - F_X(x)F_Y(y)]^2 \, dF(x, y).$$

Properties:

- $D \geq 0$,
- $D = 0$ if and only if X and Y are independent,
- D detects all types of dependence (not only monotone).

Thus D measures global deviation from independence.

5.3 Sample Version: Hoeffding's D_n

Given observations (X_i, Y_i), $i = 1, \ldots, n$, define ranks:

$$R_i = \text{rank of } X_i, \qquad S_i = \text{rank of } Y_i.$$

Hoeffding derived a U-statistic estimator of D:

$$D_n = \frac{1}{n(n-1)(n-2)(n-3)(n-4)} \sum h(Z_{i_1}, \ldots, Z_{i_5}),$$

where h is a symmetric kernel of order 5.

5.4 Hoeffding's D_n (computational form; continuous case)

`https://arxiv.org/pdf/2010.09712`

For each observation (X_i, Y_i), define the counts of other points in the four open quadrants around it:

$$a_i = \#\{\, j : \ X_j < X_i,\ Y_j > Y_i \,\}, \qquad b_i = \#\{\, j : \ X_j > X_i,\ Y_j > Y_i \,\},$$
$$c_i = \#\{\, j : \ X_j < X_i,\ Y_j < Y_i \,\}, \qquad d_i = \#\{\, j : \ X_j > X_i,\ Y_j < Y_i \,\}.$$

(Here c_i counts the number of observations lying strictly below-left of (X_i, Y_i).)

In the continuous case (no ties), Hoeffding's statistic can be computed as

$$D_n = \frac{1}{n(n-1)(n-2)(n-3)(n-4)} \sum_{i=1}^{n} \Big[a_i(a_i-1)d_i(d_i-1) + b_i(b_i-1)c_i(c_i-1) - 2a_ib_ic_id_i \Big].$$

5.5 Another version

`https://www.sciencedirect.com/science/article/pii/S1572312707000408`

Hoeffding's test is based on the fact that

$$D(x,y) \;=\; F_{X,Y}(x,y) - F_X(x)F_Y(y) \;=\; 0 \quad \forall (x,y)$$

if and only if X and Y are independent. An empirical estimator of the functional $\int D^2(x,y)\, dF(x,y)$ leads to Hoeffding's statistic D_n.

Given i.i.d. observations (X_i, Y_i), $i = 1, \dots, n$, let

$$R_i = \text{rank of } X_i \text{ among } X_1, \dots, X_n, \qquad S_i = \text{rank of } Y_i \text{ among } Y_1, \dots, Y_n.$$

Define

$$c_i = \#\Big\{\, j : \ X_j \le X_i \text{ and } Y_j \le Y_i \Big\}.$$

(For continuous data with no ties, $\le$ can be replaced by $<$ without changing the value.)

Define the three rank-sums

$$Q = \sum_{i=1}^{n} (R_i - 1)(R_i - 2)(S_i - 1)(S_i - 2),$$

$$R = \sum_{i=1}^{n} (R_i - 2)(S_i - 2)\, c_i,$$

$$S = \sum_{i=1}^{n} (c_i - 1)c_i.$$

Then Hoeffding's statistic is

$$D_n = \frac{Q - 2(n-2)R + (n-2)(n-3)S}{n(n-1)(n-2)(n-3)(n-4)}.$$

Decision rule (typical): Large values of D_n provide evidence against $H_0 : X \perp Y$. In practice, p-values are obtained from tables, asymptotic approximations, or permutation/Monte Carlo calibration under H_0.

5.6 Key Properties

- **Distribution-free under H_0:** The null distribution depends only on ranks.
- **Consistency:** $D_n \to 0$ under independence, and $D_n > 0$ for any fixed dependence structure.
- **U-statistic structure:** D_n is a U-statistic of order 5, so asymptotic normality follows from U-statistic theory.
- **Broad alternatives:** Unlike Kendall or Spearman, Hoeffding's test detects non-monotone relationships.

5.7 Comparison with Other Rank Tests

Test	Detects	Based on	Sensitivity
Pearson	General linear	Covariance	Linear only
Spearman	Monotone	Rank covariance	Monotone
Kendall	Monotone	Concordance probability	Monotone
Hoeffding	General	CDF difference	Any dependence

5.8 Interpretation

Hoeffding's statistic measures how far the empirical joint distribution deviates from the product of empirical marginals.

It is therefore:

- nonparametric,
- sensitive to broad alternatives,
- computationally heavier than Kendall or Spearman.

In modern terminology, Hoeffding's test is an early example of measuring dependence through distributional discrepancy, anticipating later ideas such as:

- distance covariance,
- kernel-based dependence measures,
- copula-based methods.

5.9 When Can Hoeffding's Test Perform Poorly?

Hoeffding's test is consistent against all fixed alternatives. That is, if X and Y are dependent, then

$$D_n \longrightarrow D > 0.$$

However, consistency does not guarantee high power in finite samples, nor does it imply optimality against specific alternatives.

There are several situations in which Hoeffding's test may perform poorly.

Purely monotone alternatives.

If the true relationship is strictly monotone (for example, linear, or some power), rank-based measures such as Kendall's τ or Spearman's ρ_s are typically more powerful. Hoeffding's test is designed to detect all types of departures from independence, so its power is spread over a broad class of alternatives. As a result, it can lose efficiency against simple monotone alternatives.

Small sample sizes.

Hoeffding's statistic is a U-statistic of order five. Higher-order U-statistics generally have larger variance, so larger samples are needed for stable behavior. In small samples, the test may therefore have low power.

Very weak local alternatives.

Under local alternatives approaching independence, Hoeffding's test may have lower Pitman efficiency (it is an asymptotic measure for comparing two tests, where a lower value means less useful than the alternative) than tests tailored to specific alternatives (such as Pearson's correlation for linear dependence).

Discrete data and ties.

The classical theory assumes continuous distributions. When many ties are present:

- the null distribution changes,
- variance formulas require adjustment,
- power can deteriorate.

Summary.

Hoeffding's test is:

- consistent against all alternatives,
- broadly sensitive to nonlinear dependence,
- but not necessarily optimal for specific structured alternatives.

In practice:

- Use Pearson for linear dependence,
- Use Kendall or Spearman for monotone alternatives,
- Use Hoeffding when dependence may be nonlinear or non-monotone.

There also exist other rank-based methods to test independence, and can be found in the R package `XICOR` with the paper in `https://www.tandfonline.com/doi/full/10.1080/01621459.2020.1758115`.

6 R code illustration for association methods

```
set.seed(1)
n <- 150

##  Discrete association: contingency table + chi-square
x_cat <- factor(sample(c("Low","Med","High"), n, replace=TRUE, prob=c(.3,.4,.3)))
y_cat <- factor(ifelse(runif(n) < c(Low=.25, Med=.45, High=.70)[x_cat], "Yes", "No"))
tab <- table(x_cat, y_cat)
tab
chisq.test(tab)

##  Continuous: Pearson/Spearman/Kendall
x <- rnorm(n)
y <- 0.8*x + rnorm(n, sd=0.8)

c(Pearson=cor(x,y,method="pearson"),
  Spearman=cor(x,y,method="spearman"),
  Kendall=cor(x,y,method="kendall"))

cor.test(x, y, method="pearson")
cor.test(x, y, method="spearman", exact=FALSE)
cor.test(x, y, method="kendall", exact=FALSE)

##  Independence => Cov=0 (simulation check)
x0 <- rnorm(n); y0 <- rnorm(n)
c(cov=cov(x0,y0), r=cor(x0,y0))

##  Zero correlation does NOT imply independence: Y = X^2
x1 <- rnorm(n)
y1 <- x1^2
c(cov=cov(x1,y1), r=cor(x1,y1))

##  Bivariate normal: rho=0 <=> independence (illustration)
Sigma0 <- matrix(c(1,0,0,1), 2, 2)
xy_ind <- MASS::mvrnorm(n, mu=c(0,0), Sigma=Sigma0)
cor(xy_ind[,1], xy_ind[,2])

Sigma_r <- matrix(c(1,0.6,0.6,1), 2, 2)
xy_dep <- MASS::mvrnorm(n, mu=c(0,0), Sigma=Sigma_r)
cor(xy_dep[,1], xy_dep[,2])

##  Correlation <-> simple linear regression slope
fit <- lm(y ~ x)
b1 <- coef(fit)[2]
r  <- cor(x,y)
sx <- sd(x); sy <- sd(y)
```

```
c(beta1_hat=b1, r_scaled=r*(sy/sx))

## t-test for Pearson (matches cor.test under normality)
t_stat <- r*sqrt((n-2)/(1-r^2))
c(t_stat=t_stat, df=n-2)

##  Permutation test for correlation (distribution-free)
B <- 3000
r_obs <- cor(x,y)
r_perm <- replicate(B, cor(x, sample(y)))
p_perm <- mean(abs(r_perm) >= abs(r_obs))
p_perm

##  Bootstrap CI for correlation (resample pairs)
boot_r <- replicate(B, {
  ii <- sample.int(n, replace=TRUE)
  cor(x[ii], y[ii])
})
quantile(boot_r, c(.025,.975))

##  Multiple regression: partial correlation via residuals
p <- 3
X2 <- matrix(rnorm(n*(p-1)), n, p-1)
x1 <- 0.7*X2[,1] + rnorm(n, sd=0.7)
y2 <- 0.9*x1 + 0.5*X2[,1] - 0.4*X2[,2] + rnorm(n)

fit_full <- lm(y2 ~ x1 + X2)
summary(fit_full)

e_y  <- resid(lm(y2 ~ X2))
e_x1 <- resid(lm(x1 ~ X2))
r_partial <- cor(e_y, e_x1)
r_partial

## partial-correlation t-statistic (tests beta1=0)
t_pc <- r_partial*sqrt((n - p - 1)/(1 - r_partial^2))
c(t_partial=t_pc, df=n-p-1)

##  Precision-matrix partial correlation (Gaussian interpretation)
S <- cov(cbind(y2, x1, X2))
Omega <- solve(S)
pc_from_omega <- -Omega[1,2]/sqrt(Omega[1,1]*Omega[2,2])
c(partial_residual=r_partial, partial_precision=pc_from_omega)

##  Kendall's tau: concordant - discordant (definition check)
kendall_tau_manual <- function(x,y){
  n <- length(x)
  s <- 0
```

```
  for(i in 1:(n-1)){
    for(j in (i+1):n){
      s <- s + sign((x[j]-x[i])*(y[j]-y[i]))
    }
  }
  2*s/(n*(n-1))
}
c(tau_builtin=cor(x,y,method="kendall"),
  tau_manual=kendall_tau_manual(x,y))

##  Spearman: Pearson correlation of ranks
rx <- rank(x); ry <- rank(y)
c(rs_builtin=cor(x,y,method="spearman"),
  rs_ranks=cor(rx,ry))

##  Hoeffding's D (broad, non-monotone alternatives)
## Uses 'Hmisc' implementation: hoeffd(x,y)$D
if(!requireNamespace("Hmisc", quietly=TRUE)) install.packages("Hmisc")
Hmisc::hoeffd(x, y)$D

## Example: non-monotone dependence where Pearson may be near 0 (try multiple replications)
x2 <- runif(n, -1, 1)
y2nm <- x2^2 + rnorm(n, sd=0.01)
c(Pearson=cor(x2,y2nm),
  Spearman=cor(x2,y2nm,method="spearman"),
  Kendall=cor(x2,y2nm,method="kendall"),
  Hoeffding=Hmisc::hoeffd(x2,y2nm)$D[1,2])

##  Outlier sensitivity: Pearson vs ranks
x3 <- rnorm(n); y3 <- x3 + rnorm(n, sd=0.7)
#Before
c(Pearson=cor(x3,y3),
  Spearman=cor(x3,y3,method="spearman"),
  Kendall=cor(x3,y3,method="kendall"))

#After
y3[1] <- y3[1] + 15
c(Pearson=cor(x3,y3),
  Spearman=cor(x3,y3,method="spearman"),
  Kendall=cor(x3,y3,method="kendall"))
```

7 Rank-Based Regression Analysis

7.1 Why Regression Can Be Rank-Based

Classical linear regression is typically developed under assumptions such as:

- normality of the errors,

- constant variance (homoscedasticity),
- optimality of least squares under squared-error loss.

Under these assumptions, least squares estimators are efficient, and inference is based on t- and F-distributions.

However, in many real datasets:

- errors may be skewed or heavy-tailed,
- outliers may distort least squares estimates,
- variances may be unequal,
- second moments may not even exist.

Since least squares minimizes

$$\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2,$$

it gives high weight to large residuals, making it sensitive to extreme observations.

Rank-based regression replaces squared magnitudes with ordering information. Instead of relying on means and variances, it uses signs and ranks of residuals.

This leads to:

- robustness to outliers,
- validity under broad error distributions,
- interpretation in terms of medians rather than means,
- procedures that remain meaningful without finite variance.

Thus regression need not be tied to normality or squared-error geometry.

7.2 Two-Sample Problem as a Linear Model

The classical two-sample location problem can be written as a simple regression model.

Suppose we observe two independent samples:

$$X_1, \ldots, X_m \quad \text{(treatment)}$$

and

$$Y_1, \ldots, Y_n \quad \text{(control)}.$$

Combine them into a single response vector:

$$Z = (X_1, \ldots, X_m, Y_1, \ldots, Y_n)^T, \qquad N = m + n.$$

Define a group indicator variable

$$g = (\underbrace{1, \ldots, 1}_{m}, \underbrace{0, \ldots, 0}_{n})^T.$$

Then the model becomes

$$Z_i = \beta_0 + \Delta g_i + e_i, \qquad i = 1, \ldots, N.$$

Here:

- β_0 represents the baseline location (control group),
- Δ represents the shift between treatment and control,
- e_i are i.i.d. errors from a common distribution F.

Testing whether the two groups differ is equivalent to testing

$$H_0 : \; \Delta = 0.$$

Thus the two-sample problem is simply a regression problem with a binary predictor.

Rank-based procedures such as the Wilcoxon rank-sum test can therefore be interpreted as regression tests for Δ that do not rely on normality.

7.3 Simple Linear Regression (Median-Based Interpretation)

Now consider the simple regression model

$$Y_i = \alpha + \beta X_i + e_i,$$

with the assumption

$$\text{median}(e_i) = 0.$$

This assumption is strictly weaker than

$$\mathbb{E}(e_i) = 0,$$

since it does not require existence of moments.

Under this model,

$$\text{median}(Y_i \mid X_i = x) = \alpha + \beta x.$$

Thus β describes how the conditional median of Y changes with X.

This interpretation contrasts with classical regression, where β describes the change in the conditional mean.

7.4 Rank-based regression

Rank-based regression methods estimate α and β by enforcing this median-zero condition using order information.

They do so in three closely related ways:

- **Signs of residuals:** If the fitted slope is correct, residuals should be equally likely to be positive or negative. Sign-based tests check whether residual signs are balanced. This directly enforces the median condition.

- **Ranks of residuals:** Instead of using squared residual magnitudes, rank-based methods use the ordering of residuals. This reduces sensitivity to extreme observations while preserving information about systematic trends.
- **Medians of pairwise slopes (Theil–Sen):** Each pair of points determines a slope

$$S_{ij} = \frac{Y_j - Y_i}{X_j - X_i}.$$

  The true slope is the value that makes positive and negative pairwise deviations balance. Taking the median of $\{S_{ij}\}$ therefore enforces a global median-zero condition across all pairs.

In contrast, least squares enforces

$$\mathbb{E}(e_i) = 0,$$

which targets the conditional mean and depends heavily on squared magnitudes.

Key distinction:

- Least squares is mean-based and magnitude-driven.
- Rank-based regression is median-based and order-driven.

Because medians are resistant to extreme values, rank-based regression naturally provides robustness while retaining meaningful interpretation.

Rank-based regression methods are therefore naturally aligned with median-based modeling. They estimate slopes and intercepts using:

- signs of residuals,
- ranks of residuals,
- or medians of pairwise slopes (Theil–Sen).

These approaches remain valid under heavy tails, skewness, and mild heteroscedasticity.

In summary:

- Least squares is mean-based and variance-driven.
- Rank-based regression is median-based and order-driven.
- The latter often provides greater robustness in practice.

8 Testing Slope Based on Signs: Theil (1950)

8.1 Hypotheses and Residual-Like Quantities

Consider the simple regression model

$$Y_i = \alpha + \beta X_i + e_i, \qquad i = 1, \dots, n,$$

where the errors satisfy $\text{median}(e_i) = 0$.

We wish to test the hypothesis

$$H_0: \ \beta = \beta_0.$$

To assess this, define the adjusted quantities

$$D_i = Y_i - \beta_0 X_i.$$

If H_0 is true, then

$$D_i = \alpha + e_i.$$

Thus under H_0, all D_i share a common median α, and no systematic relationship should remain between D_i and X_i.

Therefore, testing H_0 reduces to determining whether the adjusted values D_i show any monotone trend with X_i. If they do, then the proposed slope β_0 is incorrect.

8.2 Theil Sign Statistic

We first order the data with respect to X such that $X_1 < X_2 < \cdots < X_n$.

The test statistic is

$$C = \sum_{i<j} \mathrm{sign}(D_j - D_i).$$

(if you do not order, then just use $C = \sum_{i<j} \mathrm{sign}(D_j - D_i)(X_j - X_i)$. So it is basically Kendall's tau statistic between D and X.)

To understand this statistic, note that

$$D_j - D_i = (Y_j - Y_i) - \beta_0(X_j - X_i).$$

Interpretation:

- If $\beta = \beta_0$, then the differences $D_j - D_i$ should be equally likely to be positive or negative. Thus C fluctuates around 0.
- If $\beta > \beta_0$, then $(Y_j - Y_i)$ tends to exceed $\beta_0(X_j - X_i)$, so $D_j - D_i (= (\beta - \beta_0)(X_j - X_i) + \text{Error})$ tends to be positive, and C becomes positive.
- If $\beta < \beta_0$, then $D_j - D_i$ tends to be negative, and C becomes negative.

Hence C behaves like a Kendall-type statistic applied to slope-adjusted data. It tests whether the proposed slope removes all monotone associations.

Under mild conditions, a standardized version of C is asymptotically normal.

8.3 Theil–Sen Slope Estimator

Rather than testing a fixed β_0, we can estimate β directly using pairwise slopes.

For each pair $i < j$ with $X_j \neq X_i$, define

$$S_{ij} = \frac{Y_j - Y_i}{X_j - X_i}.$$

These are all slopes determined by pairs of points.

The Theil–Sen estimator is defined as

$$\hat{\beta} = \text{median}\{S_{ij}\}.$$

Why use the median?

- It is resistant to extreme pairwise slopes.
- It is robust under heavy-tailed error distributions.
- It requires no moment assumptions.

Unlike least squares, which minimizes squared residuals, the Theil–Sen estimator uses order structure, making it highly robust.

Its breakdown point is approximately 29%, substantially higher than that of least squares.

8.4 Rank-Based Intercept Estimation

Once $\hat{\beta}$ is obtained, define residual-like quantities

$$A_i = Y_i - \hat{\beta} X_i.$$

The intercept is estimated by

$$\hat{\alpha} = \text{median}(A_1, \ldots, A_n).$$

Thus both slope and intercept are based on medians, leading to a fully robust regression fit.

8.5 Example: Cloud Seeding and Rainfall (Smith, 1967)

```
Table9.1 = data.frame(
  x.years.seeded = c(1,2,3,4,5),
  Y.double.ratio = c(1.26,1.27,1.12,1.16,1.03)
)

library(RobustLinearReg)

out <- theil_sen_regression(Y.double.ratio~x.years.seeded, data =
    Table9.1)
summary(out)
median(out$residuals)


##Comparing to OLS
```

```
outlm <- lm(Y.double.ratio~x.years.seeded, data = Table9.1)
summary(outlm)
median(outlm$residuals)
mean(outlm$residuals)
```

Here median$(\{\hat{e}_i\}) = 0$ is specifically maintained by construction. We see later that the least absolute deviations regression (also called median regression) satisfies this condition approximately, similar to the OLS, where mean$(\{\hat{e}_i\}) = 0$ is satisfied approximately.

8.6 Confidence Interval for the Slope

A rank-based confidence interval can be constructed from the ordered pairwise slopes.

```
theil.output = theil(Table9.1$x.years.seeded,
                     Table9.1$Y.double.ratio,
                     beta.0 = 0,
                     slopes = TRUE,
                     type = "t",
                     doplot = FALSE,
                     alpha = .05)

c(theil.output$L, theil.output$U)
```

Example:

$$(L, U) = (-0.15,\ 0.04).$$

Since the interval contains 0, there is no strong evidence of a systematic change in rainfall with years of seeding in this small sample.

9 Rank-Based Multiple Linear Regression

9.1 Model and Assumptions

Consider the linear regression model

$$Y_i = \mu + \boldsymbol{x}_i^T \boldsymbol{\beta} + e_i, \qquad i = 1, \ldots, n,$$

where

- $x_i \in \mathbb{R}^p$ is the predictor vector,
- $\boldsymbol{\beta} \in \mathbb{R}^p$ is the slope parameter,
- e_i are independent errors satisfying

  $$\text{median}(e_i) = 0,$$

  with otherwise unspecified distribution.

Thus

$$\text{median}(Y_i \mid x_i) = \mu + \boldsymbol{x}_i^T \boldsymbol{\beta}.$$

Unlike least squares regression, no assumptions are made about:

- normality,
- homoscedasticity,
- finite second moments.

Inference relies only on ordering of residuals.

9.2 Rank-Based Objective Function

For a given $\boldsymbol{\beta}$, define residuals (ignoring μ and it makes sense due to the explaination given the next subsection)

$$r_i(\boldsymbol{\beta}) = Y_i - x_i^T \boldsymbol{\beta}.$$

Let $R_i(\boldsymbol{\beta})$ denote the rank (or midrank) of $r_i(\boldsymbol{\beta})$ among $\{r_1(\boldsymbol{\beta}), \ldots, r_n(\boldsymbol{\beta})\}$.
Define centered ranks

$$R_i^c(\boldsymbol{\beta}) = R_i(\boldsymbol{\beta}) - \frac{n+1}{2}.$$

The rank regression estimator $\hat{\boldsymbol{\beta}}$ minimizes the dispersion function

$$D(\boldsymbol{\beta}) = \sum_{i=1}^{n} R_i^c(\boldsymbol{\beta})\, r_i(\boldsymbol{\beta}).$$

This criterion depends only on residual ordering. Large residual magnitudes do not dominate unless they are extreme in rank.

9.3 Equivalence of $D(\boldsymbol{\beta})$ and $D(\boldsymbol{\beta}, \mu)$

For the multiple regression model

$$Y_i = \mu + \boldsymbol{x}_i^T \boldsymbol{\beta} + e_i,$$

it is natural to write the rank-based dispersion as a function of both the intercept μ and slope $\boldsymbol{\beta}$:

$$D(\boldsymbol{\beta}, \mu) = \sum_{i=1}^{n} R_i^c(\boldsymbol{\beta}, \mu)\, r_i(\boldsymbol{\beta}, \mu), \qquad r_i(\boldsymbol{\beta}, \mu) = Y_i - \mu - \boldsymbol{x}_i^T \boldsymbol{\beta},$$

where $R_i(\boldsymbol{\beta}, \mu)$ is the rank (or midrank) of $r_i(\boldsymbol{\beta}, \mu)$ among $\{r_1(\boldsymbol{\beta}, \mu), \ldots, r_n(\boldsymbol{\beta}, \mu)\}$ and

$$R_i^c(\boldsymbol{\beta}, \mu) = R_i(\boldsymbol{\beta}, \mu) - \frac{n+1}{2}.$$

Key observation (translation invariance of ranks). For any constant $c \in \mathbb{R}$, the residual vectors satisfy

$$r_i(\boldsymbol{\beta}, \mu + c) = r_i(\boldsymbol{\beta}, \mu) - c, \qquad i = 1, \ldots, n.$$

Subtracting the same constant from every residual does not change their relative ordering. Hence the ranks are unchanged:

$$R_i(\boldsymbol{\beta}, \mu + c) = R_i(\boldsymbol{\beta}, \mu), \qquad R_i^c(\boldsymbol{\beta}, \mu + c) = R_i^c(\boldsymbol{\beta}, \mu).$$

Therefore,

$$D(\boldsymbol{\beta}, \mu + c) = \sum_{i=1}^{n} R_i^c(\boldsymbol{\beta}, \mu)\, \{r_i(\boldsymbol{\beta}, \mu) - c\} = D(\boldsymbol{\beta}, \mu) - c \sum_{i=1}^{n} R_i^c(\boldsymbol{\beta}, \mu).$$

Centered ranks sum to zero. Since $\sum_{i=1}^{n} R_i(\boldsymbol{\beta}, \mu) = 1 + \cdots + n = \frac{n(n+1)}{2}$, we have

$$\sum_{i=1}^{n} R_i^c(\boldsymbol{\beta}, \mu) = \sum_{i=1}^{n} \left(R_i(\boldsymbol{\beta}, \mu) - \frac{n+1}{2} \right) = \frac{n(n+1)}{2} - n \cdot \frac{n+1}{2} = 0.$$

Hence the second term vanishes and we obtain

$$D(\boldsymbol{\beta}, \mu + c) = D(\boldsymbol{\beta}, \mu) \qquad \text{for all } c \in \mathbb{R}.$$

Conclusion (equivalence). The dispersion does not depend on μ:

$$D(\boldsymbol{\beta}, \mu) = D(\boldsymbol{\beta}, 0).$$

Therefore we may define

$$D(\boldsymbol{\beta}) \equiv D(\boldsymbol{\beta}, 0),$$

and minimizing over $(\boldsymbol{\beta}, \mu)$ is equivalent to minimizing over $\boldsymbol{\beta}$ only:

$$\arg\min_{\boldsymbol{\beta}, \mu} D(\boldsymbol{\beta}, \mu) = \Big\{ (\boldsymbol{\beta}, \mu) : \ \boldsymbol{\beta} \in \arg\min_{\boldsymbol{\beta}} D(\boldsymbol{\beta}) \Big\}.$$

In practice, one estimates $\boldsymbol{\beta}$ by minimizing $D(\boldsymbol{\beta})$, and then estimates μ separately using the median residual condition

$$\hat{\mu} = \text{median}\{Y_i - \boldsymbol{x}_i^T \hat{\boldsymbol{\beta}}\}.$$

9.4 Convexity and Optimization

The function $D(\boldsymbol{\beta})$ is:

- continuous,
- piecewise linear,
- convex in $\boldsymbol{\beta}$.

Convexity ensures that:

- a global minimum exists,
- numerical optimization is stable,
- linear programming techniques may be used.

In higher dimensions, slope changes occur when residuals exchange order, which corresponds to hyperplanes

$$Y_i - \boldsymbol{x}_i^T \boldsymbol{\beta} = Y_j - \boldsymbol{x}_j^T \boldsymbol{\beta}.$$

These hyperplanes partition $\mathbb{R}^p$ into polyhedral regions where ranks remain fixed. Within each region, $D(\boldsymbol{\beta})$ is linear.

9.5 Multiple Linear Rank Regression: Jaeckel and Hettmansperger–McKean Framework

Consider the linear model

$$Y_i = \mu + \boldsymbol{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \qquad i = 1, \dots, n,$$

with $\boldsymbol{x}_i \in \mathbb{R}^p$ and no parametric assumption on ε_i except continuity.

Define residuals

$$r_i(\boldsymbol{\beta}, \mu) = Y_i - \mu - \boldsymbol{x}_i^T \boldsymbol{\beta},$$

and let $R_i(\boldsymbol{\beta}, \mu)$ denote the rank (or midrank) of $r_i(\boldsymbol{\beta}, \mu)$ among $\{r_1, \dots, r_n\}$.

1. Jaeckel's Dispersion Function (1972)

For a score function $a(\cdot)$ define

$$D_J(\boldsymbol{\beta}, \mu) = \sum_{i=1}^{n} a(R_i(\boldsymbol{\beta}, \mu))\; r_i(\boldsymbol{\beta}, \mu).$$

The rank regression estimator $(\hat{\boldsymbol{\beta}}, \hat{\mu})$ minimizes D_J.

Estimating equations

$$\sum_{i=1}^{n} a(R_i(\hat{\boldsymbol{\beta}}, \hat{\mu})) = 0, \qquad \sum_{i=1}^{n} \boldsymbol{x}_i \, a(R_i(\hat{\boldsymbol{\beta}}, \hat{\mu})) = 0.$$

Thus rank regression projects rank scores onto the column space of X.

2. Centered-Rank (Wilcoxon) Dispersion

Define centered ranks

$$R_i^c(\boldsymbol{\beta}, \mu) = R_i(\boldsymbol{\beta}, \mu) - \frac{n+1}{2}.$$

Then

$$D_W(\boldsymbol{\beta}, \mu) = \sum_{i=1}^{n} R_i^c(\boldsymbol{\beta}, \mu)\, r_i(\boldsymbol{\beta}, \mu).$$

This equals Jaeckel's dispersion with Wilcoxon scores

$$a(R) = R - \frac{n+1}{2}.$$

Hence

$$D_W(\boldsymbol{\beta}, \mu) = D_J(\boldsymbol{\beta}, \mu) \text{ with Wilcoxon scores.}$$

Since $\sum_i R_i^c = 0$, location equivariance holds automatically.

3. Common Score Choices

Let the function a be $a(r) = f(r/(n+1))$ a score function on $(0,1)$.

- **Wilcoxon:**
$$f(u) = u - \tfrac{1}{2}$$

- **Normal (van der Waerden):**
$$f(u) = \Phi^{-1}(u),$$
where Φ is the standard normal CDF and Φ^{-1} is the `qnorm` function in R.

- **Sign (LAD-type):**
$$f(u) = \mathrm{sign}(u - \tfrac{1}{2})$$

- **Logistic:**
$$f(u) = \log\left(\frac{u}{1-u}\right)$$

Different scores yield different efficiency–robustness tradeoffs.

Thus, rank regression is nearly as efficient as OLS under normality and can be substantially more efficient under heavy tails.

Example: LAD Regression as Sign-Score Rank Regression

Consider the linear model

$$Y_i = \boldsymbol{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \qquad r_i(\boldsymbol{\beta}) = Y_i - \boldsymbol{x}_i^T \boldsymbol{\beta}.$$

LAD regression. The least absolute deviations estimator solves

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} |r_i(\boldsymbol{\beta})|.$$

The first-order condition (for $r_i \neq 0$) is

$$\sum_{i=1}^{n} \boldsymbol{x}_i \ \mathrm{sign}\Big(r_i(\hat{\boldsymbol{\beta}})\Big) = 0.$$

Rank regression. Jaeckel's dispersion is

$$D(\boldsymbol{\beta}) = \sum_{i=1}^{n} a(R_i(\boldsymbol{\beta})) \ r_i(\boldsymbol{\beta}),$$

where $R_i(\boldsymbol{\beta})$ is the rank of $r_i(\boldsymbol{\beta})$.

The corresponding estimating equation is

$$\sum_{i=1}^{n} \boldsymbol{x}_i \, a\Big(R_i(\hat{\boldsymbol{\beta}})\Big) = 0.$$

Sign scores. If we choose the score function

$$a(r) = \operatorname{sign}(r/(n+1) - \tfrac{1}{2}),$$

This means: 1) If residual rank is above median $\rightarrow$ score $= +1$ and 2) If residual rank is below median $\rightarrow$ score $= -1$.

Then the sign of the centered rank is the same as the sign of the residual: $\operatorname{sign}(R_i/(n+1) - \frac{1}{2}) = \operatorname{sign}(r_i)$

Thus,

$$a(R_i(\boldsymbol{\beta})) = \operatorname{sign}(r_i(\boldsymbol{\beta}))\,.$$

Hence the rank-regression estimating equation becomes

$$\sum_{i=1}^{n} \boldsymbol{x}_i \operatorname{sign}\Big(r_i(\hat{\boldsymbol{\beta}})\Big) = 0,$$

which is exactly the LAD estimating equation.

LAD regression is rank regression with sign scores.

4. Hettmansperger–McKean Development (1977–2011)

Hettmansperger and McKean extended Jaeckel's theory by:

1. Establishing asymptotic normality;
2. Deriving semiparametric efficiency bounds;
3. Developing general linear hypothesis testing;
4. Providing computational algorithms.

Reference: Hettmansperger and McKean (2011), Robust Nonparametric Statistical Methods.

5. Asymptotic Distribution

Under regularity conditions,

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \ \xrightarrow{d}\ N\big(0, \tau^2 (X^T X)^{-1}\big)\,,$$

where

$$\tau^2 = \frac{\int_0^1 a^2(u)\,du}{\Big(\int_0^1 a'(u) f(F^{-1}(u))\,du\Big)^2}.$$

Normal scores achieve semiparametric efficiency under Gaussian errors.

6. General Linear Hypothesis

Test

$$H_0 : M\boldsymbol{\beta} = 0.$$

Let D_F and D_R be the full and reduced dispersions.
Define reduction in dispersion

$$RD = D_R - D_F.$$

Under H_0,

$$RD \xrightarrow{d} \chi^2_q.$$

This parallels the extra sum-of-squares F-test in OLS.

7. Connections

- Wilcoxon rank-sum test: special case when $p = 1$ and x_i is a group indicator.
- LAD regression: sign-score rank regression.
- Quantile regression: minimizes asymmetric absolute loss (different objective).
- M-estimation: smooth analog of rank-based estimating equations.
- Permutation tests: asymptotically equivalent under symmetry.
- Jaeckel provides the general score-based dispersion framework.
- Hettmansperger–McKean provide asymptotics and inference.
- Centered-rank regression is the Wilcoxon special case.

9.6 General Linear Hypothesis

Let M be a $q \times p$ full row rank matrix.
We test

$$H_0 : \boldsymbol{M\beta} = 0 \qquad \text{vs} \qquad H_A : \boldsymbol{M\beta} \neq 0.$$

Let $\Omega_F = \text{col}(X)$ denote the full model space, and let Ω_R denote the reduced space under H_0.
Compute the rank-based estimators:

$$\hat{\boldsymbol{\beta}}_F \quad \text{and} \quad \hat{\boldsymbol{\beta}}_R.$$

Define the corresponding dispersions:

$$D_F = D(\hat{\boldsymbol{\beta}}_F), \qquad D_R = D(\hat{\boldsymbol{\beta}}_R).$$

The reduction in dispersion is

$$RD = D_R - D_F.$$

Large values of RD provide evidence against H_0.
This plays the same role as the partial F-statistic in least squares regression, but is based on rank-score geometry rather than squared errors.

```r
# ------------------------------------------------------------
# Example: Rank-based test of H0: beta2 = beta3 = 0
# ------------------------------------------------------------

set.seed(123)

# -------------------------
# 1. Simulate data
# -------------------------
n <- 120

x1 <- rnorm(n)
x2 <- rnorm(n)
x3 <- rnorm(n)

# Heavy-tailed errors to emphasize robustness
e  <- rt(n, df = 3)

# True model: beta2 != 0, beta3 = 0
y <- 2 + 1.5*x1 + 1.0*x2 + 0*x3 + e

dat <- data.frame(y, x1, x2, x3)

# -------------------------
# 2. Fit rank-based models
# -------------------------
library(Rfit)

# Full model
fitF <- rfit(y ~ x1 + x2 + x3, data = dat)

# Reduced model (drop x2 and x3)
fitR <- rfit(y ~ x1, data = dat)

# -------------------------
# 3. Rank-based ANOVA-type test
# -------------------------
anova_result <- drop.test(fitF, fitR)

anova_result
```

Additional examples and details: https://journal.r-project.org/articles/RJ-2012-014/ and https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=1347&context=r-journal

10 Geometric Interpretation

Least squares solves
$$\min_{\boldsymbol{\eta}\in\Omega_F} \|\boldsymbol{Y}-\boldsymbol{\eta}\|_2^2.$$

Rank regression solves
$$\min_{\boldsymbol{\eta}\in\Omega_F} \sum_{i=1}^{n} R_i^c(\boldsymbol{\eta})\,(Y_i-\eta_i).$$

Thus:

- Least squares uses Euclidean projection.
- Rank regression uses projection under a rank-score functional.

The model space Ω_F remains a linear subspace, but distance is measured through residual ordering rather than squared magnitude.

Because the objective is convex and piecewise linear, solutions typically occur at vertices of polyhedral regions defined by residual order changes.

10.1 Bootstrap Inference

Confidence intervals for $\boldsymbol{\beta}$ may be obtained using:

1. Residual bootstrap:
$$Y_i^* = \boldsymbol{x}_i^T\hat{\boldsymbol{\beta}} + r_i^*,$$
where r_i^* are resampled residuals.
2. Pairs bootstrap: Resample $(Y_i, \boldsymbol{x}_i)$ jointly as we did in the linear model regression example during the bootstrap example illustration.

After B replications, a $(1-\alpha)$ percentile confidence interval for β_j is given by the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of $\{\hat{\beta}_j^{*(b)}\}_{b=1}^B$. Typically, $B \geq 1000$ is recommended.

Residual Bootstrap (Regression)

Consider the regression model
$$Y_i = x_i^\top\boldsymbol{\beta}_0 + e_i, \qquad i = 1,\ldots,n,$$
with i.i.d. errors e_i with $\mathbb{E}(e_i) = 0$ (or $\mathrm{median}(e_i) = 0$ for rank-based regression).

Fit the model using the full data to obtain $\hat{\boldsymbol{\beta}}$ and (estimated) residuals
$$\hat{r}_i = Y_i - x_i^\top\hat{\boldsymbol{\beta}}.$$

Center residuals (mean or median) to enforce the appropriate location constraint:
$$\tilde{r}_i = \hat{r}_i - \bar{r} \quad (\text{or } \tilde{r}_i = \hat{r}_i - \mathrm{median}(\hat{r})).$$

For $b = 1,\ldots,B$:

1. Resample $\{r_1^{*(b)},\ldots,r_n^{*(b)}\}$ with replacement from $\{\tilde{r}_1,\ldots,\tilde{r}_n\}$.

2. Generate bootstrap responses
$$Y_i^{*(b)} = x_i^\top \hat{\boldsymbol{\beta}} + r_i^{*(b)}.$$

3. Refit the model on $\{(x_i, Y_i^{*(b)})\}_{i=1}^n$ to obtain $\hat{\boldsymbol{\beta}}^{*(b)}$.

A $(1-\alpha)$ percentile confidence interval for β_j is given by the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of $\{\hat{\beta}_j^{*(b)}\}_{b=1}^B$.

11 References for this lecture

- HWC Chapter 8.
- HWC Chapter 9.1–9.4, 9.6 (rank-based regression, geometry, inference).
- Hettmansperger, Thomas P., and Joseph W. McKean. (2010). Robust Nonparametric Statistical Methods. CRC Press.